

XR-500 Industrial Controller

Safety & Standard Operating Procedure (SOP)

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1. Safety Requirements & Compliance

1.1 Personal Protective Equipment (PPE)

All operators and maintenance technicians working within 2 meters of the XR-500 controlled process skid must wear the following mandatory safety gear:

- Safety Glasses with side shields (ANSI Z87.1 approved)
- Steel-toe work boots (ASTM F2413)
- Heat-resistant and chemical-resistant gloves (EN 374 compliant)

1.2 Interlock & Safety Systems

The XR-500 is equipped with redundant dual-channel safety circuits. Under no circumstances should safety interlocks, enclosure door microswitches, or pressure relief valves be bypassed or jumpered.

2. Operational Safety Limits & Prohibited Conditions

2.1 Critical Safety Thresholds

To prevent catastrophic mechanical failure or fluid line rupture, the unit **MUST NOT** be operated outside the strict safety limits defined below.

1.	=====		
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2.	CRITICAL SAFETY OPERATING BOUNDARIES		
3.	=====		
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4.	Parameter	Absolute Safety Limit	Action On Exceedance
5.	-----		
6.	Maximum System Pressure	10.5 bar	Immediate Pressure Relief Valve trip

7. Maximum Line Flow Rate	180.0 L/min	Pump Drive Overload Trip
8. Ambient Operating Temp	-10°C to 40°C	System Thermal Lockout
9. Maximum Supply Voltage	30.0 VDC	Power Board Overvoltage Shutdown
10. =====		
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2.2 Prohibited Actions

- **NEVER** attempt to modify electrical wiring without disabling incoming DC breaker.
- **NEVER** operate the controller in environments where ambient temperature exceeds 40°C.
- **NEVER** clear fault logs while the process skid is active in **AUTO** mode.

3. Emergency Shutdown Procedure (E-STOP)

In the event of fluid leaks, abnormal noise, smoke, or uncontrolled parameter runaway:

1. **Press the nearest Physical Emergency Stop (E-Stop) Button** located on the front cabinet door or remote panel station.
2. The controller automatically drops the main isolation contactor within **< 50 ms**, removing power from all connected actuator solenoids and motor drives.
3. Close manual fluid isolation valves V-101 (Inlet) and V-102 (Outlet).
4. Verify system pressure gauge indicates **0.0 bar** before approaching the fluid manifold assembly.
5. Notify the Area Safety Supervisor and file an Incident Log Report.

4. Routine Safety Inspection Procedure

Perform weekly safety checks according to the following checklist:

1. Test Emergency Stop button responsiveness while operating in low-flow test mode.
2. Check enclosure door gaskets for signs of degradation or moisture intrusion.
3. Inspect ground bonding strap for tightness and freedom from corrosion.
4. Verify ambient enclosure temperature reading on HMI panel remains below 40°C during peak operation.

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